

SI 140A-01 Probability & Statistics for EECS, Fall 2026

Homework 2

Name: 杨涵

Student ID: 202553/064

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Read all the instructions below carefully before you start working on the assignment, and before you make a submission.

- You are required to write down all the major steps towards making your conclusions; otherwise you may obtain limited points of the problem.
- Write your homework in English; otherwise you will get no points of this homework.
- Any form of plagiarism will lead to 0 point of this homework.

Problem 1

Show that for any events A and B ,

$$P(A) + P(B) - 1 \leq P(A \cap B) \leq P(A \cup B) \leq P(A) + P(B).$$

For each of these three inequalities, give a simple criterion for when the inequality is actually an equality (e.g., give a simple condition such that $P(A \cap B) = P(A \cup B)$ if and only if the condition holds).

According to the inclusion-exclusion formula.

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Since $P(A \cup B) \leq 1$, so that $P(A) + P(B) - P(A \cap B) \leq 1 \Rightarrow P(A) + P(B) - 1 \leq P(A \cap B)$

Equality holds if and only if $P(A \cup B) = 1$

since $A \cap B \subseteq A \cup B \Rightarrow P(A \cap B) \leq P(A \cup B)$, Equality holds if and only if $P(A \cap B) = 0$

since $P(A \cap B) \geq 0 \Rightarrow P(A) + P(B) - P(A \cup B) \geq 0 \Rightarrow P(A \cup B) \leq P(A) + P(B)$

Equality holds if and only if $P(A \cap B) = 0$

Problem 2

Let A and B be events. The *symmetric difference* $A \Delta B$ is defined to be the set of all elements that are in A or B but not both. In logic and engineering, this event is also called the XOR (*exclusive or*) of A and B . Show that

$$P(A \Delta B) = P(A) + P(B) - 2P(A \cap B),$$

directly using the axioms of probability.

By definition, $A \Delta B = (A \setminus B) \cup (B \setminus A)$

$A \setminus B$ and $B \setminus A$ are disjoint events, so that $P(A \Delta B) = P(A \setminus B) + P(B \setminus A)$

as $A = (A \setminus B) \cup (A \cap B)$. $B = (B \setminus A) \cup (A \cap B)$

$$P(A \setminus B) = P(A) - P(A \cap B), \quad P(B \setminus A) = P(B) - P(A \cap B)$$

$$\text{so that } P(A \Delta B) = P(A) - P(A \cap B) + P(B) - P(A \cap B) = P(A) + P(B) - 2P(A \cap B)$$

Problem 3

Events A and B are *independent* if $P(A \cap B) = P(A)P(B)$.

1. Give an example of independent events A and B in a finite sample space S (with neither equal to $\emptyset$ or S).
2. Consider the experiment of picking a random point in the rectangle

$$R = \{(x, y) : 0 < x < 1, 0 < y < 1\},$$

where the probability of the point being in any particular region contained within R is the area of that region. Let A_1 and B_1 be rectangles contained within R , with areas not equal to 0 or 1. Let A be the event that the random point is in A_1 , and B be the event that the random point is in B_1 . Give a geometric description of when it is true that A and B are independent. Also, give an example where they are independent and another example where they are not independent.

3. Show that if A and B are independent, then

$$P(A \cup B) = P(A) + P(B) - P(A)P(B) = 1 - P(A^c)P(B^c).$$

1. assume $S = \{1, 2, 3, 4\}$, $A = \{1, 2\}$, $B = \{2, 4\}$,

$P(A) = \frac{1}{2}$, $P(B) = \frac{1}{2}$, $A \cap B = \{2\}$, $P(A \cap B) = \frac{1}{4} = P(A) \cdot P(B)$. A and B are independent.

2. $P(A) = S_{A_1}$, $P(B) = S_{B_1}$, $P(A \cap B) = S_{A_1 \cap B_1}$,

A and B are independent when $S_{A_1} \cdot S_{B_1} = S_{A_1 \cap B_1}$

so geometrically, the overlap of A_1 and B_1 should have an area equal to the product of their individual areas.

(1) A and B are independent

let $A = \{(x, y) | 0 < x < \frac{1}{2}, 0 < y < 1\}$, $B = \{(x, y) | 0 < x < 1, 0 < y < \frac{1}{2}\}$

$$P(A) = \frac{1}{2}, P(B) = \frac{1}{2}, P(A \cap B) = \frac{1}{4} = P(A) \cdot P(B)$$

(2) A and B are not independent.

let $A = B = \{(x, y) | 0 < x < \frac{1}{2}, 0 < y < 1\}$.

$$P(A) = P(B) = \frac{1}{2}, P(A \cap B) = \frac{1}{2} \neq P(A) \cdot P(B)$$

3. A and B are independent. $P(A \cap B) = P(A) \cdot P(B)$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = P(A) + P(B) - P(A) \cdot P(B)$$

$$\Rightarrow P(A \cup B) = P(A) + P(B) - P(A) \cdot P(B) = 1 - P(A^c) \cdot P(B^c)$$

$$P(A^c) \cdot P(B^c) = P((A \cup B)^c) = 1 - P(A \cup B) \text{ , so } P(A \cup B) = 1 - P(A^c) \cdot P(B^c)$$

Problem 4

Consider four nonstandard dice (the *Efron dice*), whose sides are labeled as follows (the 6 sides on each die are equally likely).

A: 4, 4, 4, 4, 0, 0

B: 3, 3, 3, 3, 3, 3

C: 6, 6, 2, 2, 2, 2

D: 5, 5, 5, 1, 1, 1

These four dice are each rolled once. Let A be the result for die A , B be the result for die B , etc.

- Find $P(A > B)$, $P(B > C)$, $P(C > D)$, and $P(D > A)$.
- Is the event $A > B$ independent of the event $B > C$? Is the event $B > C$ independent of the event $C > D$? Explain.

1. $A > B$ happens when $A=4$, $P(A > B) = \frac{4}{6} = \frac{2}{3}$

$B > C$ happens when $C=2$, $P(B > C) = \frac{4}{6} = \frac{2}{3}$

$C > D$ happens when $\begin{cases} C=6, P_1 = \frac{2}{6} = \frac{1}{3} \\ C=2 \text{ and } D=1, P_2 = \frac{2}{6} \times \frac{1}{3} = \frac{1}{9} \end{cases} \Rightarrow P(C > D) = \frac{1}{3} + \frac{1}{9} = \frac{4}{9}$

$D > A$ happens when $\begin{cases} D=5, P_1 = \frac{1}{6} \\ D=1 \text{ and } A=0, P_2 = \frac{1}{6} \times \frac{2}{6} = \frac{1}{18} \end{cases} \Rightarrow P(D > A) = \frac{1}{6} + \frac{1}{18} = \frac{4}{9}$

2. $P(A > B) = \frac{2}{3}$, $P(B > C) = \frac{2}{3}$. $(A > B) \cap (B > C)$ when $\begin{cases} A=4 \\ B=3 \\ C=2 \end{cases} \Rightarrow P = \frac{2}{3} \times \frac{2}{3} = \frac{4}{9} = P(A > B) \cdot P(B > C)$
event $A > B$ and $B > C$ are independent.

$P(B > C) = \frac{2}{3}$, $P(C > D) = \frac{4}{9}$, $(B > C) \cap (C > D)$ when $\begin{cases} B=3 \\ C=2 \\ D=1 \end{cases} \Rightarrow P = \frac{1}{3} \times \frac{2}{3} \times \frac{1}{3} = \frac{1}{27} \neq P(B > C) \cdot P(C > D)$

event $B > C$ and $C > D$ are not independent,

because $B > C \Rightarrow C=2$, which changes the probability that $C > D$